

MAT1002 Lecture 24,

Outline

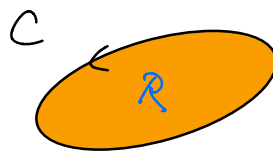
- Green's theorem
 - ↳ Idea
 - ↳ More general regions
- Parametric Surfaces
 - ↳ Tangent planes
 - ↳ Surface areas

Green's Theorem

In short :

$$\oint_C \vec{F} \cdot \vec{T} ds = \iint_R \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dA$$

$$\oint_C \vec{F} \cdot \vec{n} ds = \iint_R \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) dA$$



where C is a simple closed curve that the counterclockwise boundary of a bounded region R .

Here : $\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$: "circulation density"

$\approx$ circulation along a small curve around a point
area of bounded region

$\text{div} \vec{F}$

$\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)$: "outward) flux density" or "divergence" ✓

$\approx$ outward flux across a small curve around a point
area of bounded region

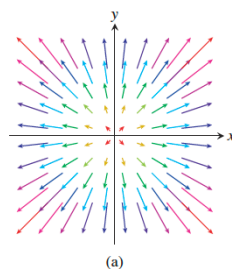
$\text{div} \vec{F}(x,y) > 0$
: expanding

< 0 : shrinking / compressing.

e.g. (a) $\vec{F} = \langle cx, cy \rangle$ (picture shows for $c > 0$)

$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$: no rotation at very small scale.

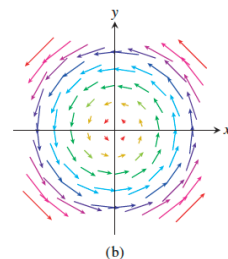
$\text{div} \vec{F} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 2c$ $\begin{cases} > 0, & \text{if } c > 0 \rightarrow \text{diverges/expands} \\ < 0, & \text{if } c < 0 \rightarrow \text{compress} \end{cases}$

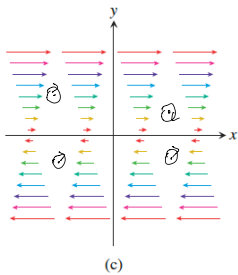


(b) $\vec{F} = \langle -cy, cx \rangle$ (picture shows for $c > 0$)

$\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = c + c = 2c$ $\begin{cases} > 0, & \text{if } c > 0 \rightarrow \text{CCW} \\ < 0, & \text{if } c < 0 \rightarrow \text{CW} \end{cases}$

$\text{div} \vec{F} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$: no expansion



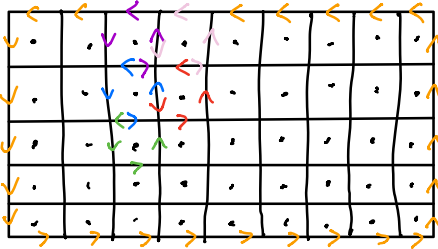


(c) $\vec{F} = \langle cy, 0 \rangle$ (picture shows for $c > 0$)

• $\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = -c$ $\begin{cases} \text{clockwise rotation at small scale, if } c > 0 \\ \text{counterclockwise " " " " , if } c < 0 \end{cases}$

• $\text{div}(\vec{F}) = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$: no expansion.

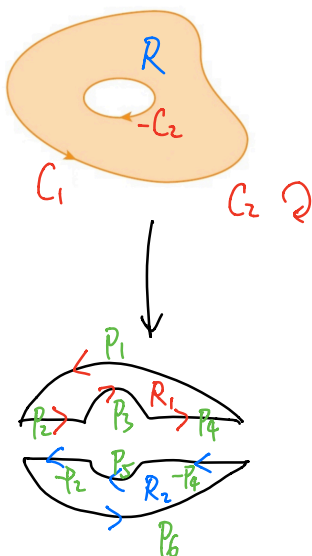
Green's Theorem in a Photo (Circulation)



- Sum up all circulations around small regions $(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y})dA$
- Adjacent "side" cancelled
- Leaving only the circulation along boundary of the whole region R .

(Idea for flux version is the same.)

More General Versions



Consider the region R on the left.

- Boundary of R is not a simple closed curve — it is the union of two such curves.
- Previous version of Green's theorem cannot be used.

• $\iint_R (N_x - M_y) dA = \iint_{R_1} + \iint_{R_2}$

↪ Need to break R first

Green's $\equiv \left(\int_{P_1} \vec{F} \cdot d\vec{r} + \int_{P_2} + \int_{P_3} + \int_{P_4} \right) + \left(\int_{-P_4} + \int_{P_5} + \int_{-P_2} + \int_{P_6} \right)$

$$= (\int_{P_1} + \int_{P_6}) + (\int_{P_3} + \int_{P_5}) = \oint_{C_1} + \oint_{C_2} = \int_{\substack{\text{Boundary} \\ \text{of } R}} \vec{F} \cdot d\vec{r}$$

Green's Theorem (General version for circulations)

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA = \int_{\substack{\text{positively oriented} \\ \text{boundary } C \text{ of } R}} \vec{F} \cdot d\vec{r}$$

↪ C is the union of one or more closed curves

(Here, C is **positively oriented** if R is always on the left of C as C being traversed.)

Parametric Surfaces

Def: A surface (in $\mathbb{R}^3$) is called a **parametric surface** if it has the form

$$\underline{x = x(u,v), \quad y = y(u,v), \quad z = z(u,v), \quad (u,v) \in R}$$

Parametric equations ↪ Parameter domain in $\mathbb{R}^2$

Hence, a parametric surface can be viewed as the range of a function

$$\vec{r} := \vec{r}(u,v) = \langle x(u,v), y(u,v), z(u,v) \rangle \text{ defined on } R.$$

Example

- (a) Find two different parametric representations for the cone given by

$$z = \sqrt{x^2 + y^2}.$$

- (b) Find a parametric representation for the sphere given by

$$x^2 + y^2 + z^2 = a^2,$$

where $a > 0$.

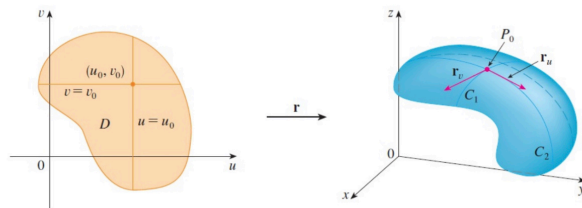
Tangent Planes to Parametric Surfaces

Let S be a parametric surface given by

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle,$$

where x , y and z have continuous partial derivatives. Let

$P_0 = \mathbf{r}(u_0, v_0)$ be a point in S . If we hold u constant by setting $u = u_0$, then $\mathbf{r}(u_0, v)$ becomes a parametrization (with parameter v) of a curve C_1 lying on S , as shown below.



A tangent vector to C_1 at P_0 is given by

$$\mathbf{r}_v := \left\langle \frac{\partial x}{\partial v}(u_0, v_0), \frac{\partial y}{\partial v}(u_0, v_0), \frac{\partial z}{\partial v}(u_0, v_0) \right\rangle.$$

Similarly, by holding v constant, $\mathbf{r}(u, v_0)$ gives a curve C_2 lying on S , and its tangent vector at P_0 is

$$\mathbf{r}_u := \left\langle \frac{\partial x}{\partial u}(u_0, v_0), \frac{\partial y}{\partial u}(u_0, v_0), \frac{\partial z}{\partial u}(u_0, v_0) \right\rangle.$$

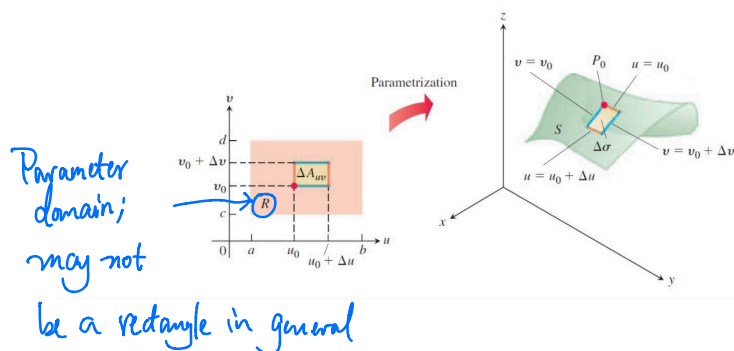
Hence, $\vec{r}_u \times \vec{r}_v$ is a normal vector to the tangent plane at a point in a parametric surface (given that $\vec{r}_u \times \vec{r}_v \neq \vec{0}$).

Def: A parametric surface S is called **Smooth** if $\vec{r}_u$ and $\vec{r}_v$ are cts, and $\vec{r}_u \times \vec{r}_v \neq \vec{0}$ for every point in the parameter domain.

Surface Areas: Parametric Forms

Let S be a smooth parametric surface. The surface area of S is equal to the sum of the areas of many small subregions, as shown below. The area $\Delta\sigma$ of each subregion is approximately the area of a parallelogram, which is

$$|\Delta u \mathbf{r}_u \times \Delta v \mathbf{r}_v| = |\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v.$$



$$\vec{r}(u_0, v_0 + \Delta v) - \vec{r}(u_0, v_0) \approx \vec{r}_v \Delta v$$

$$\begin{aligned} \vec{r}(u_0 + \Delta u, v_0) - \vec{r}(u_0, v_0) &\approx \frac{\partial \vec{r}}{\partial u}(u_0, v_0) \Delta u \\ &= \vec{r}_u \Delta u \end{aligned}$$

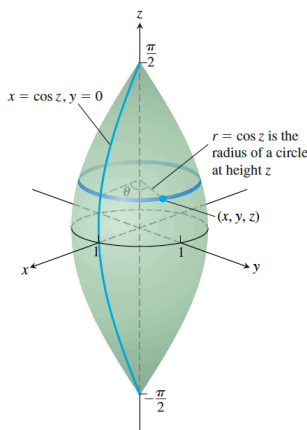
By adding terms of the form $|\mathbf{r}_u \times \mathbf{r}_v| \Delta u \Delta v$ and taking limit as $\|P\| \rightarrow 0$, we define the **surface area** of S to be $\iint_S d\sigma$, where

Area of a small piece of S

$$\iint_S d\sigma := \iint_R |\mathbf{r}_u \times \mathbf{r}_v| du dv,$$

and R is the parameter domain.

e.g. Parametrize the following surface "American football", and find its area.



Ans :

$$2\pi [\sqrt{2} + \ln(1 + \sqrt{2})].$$

Explicit form : $z = f(x, y)$

For a smooth surface S given by $z = f(x, y)$, $(x, y) \in R$, we have

$$x = u, \quad y = v, \quad z = f(u, v), \quad (u, v) \in R.$$

One can show that surface area of S is

$$A = \iint_R \sqrt{f_x^2 + f_y^2 + 1} \, dx \, dy,$$

where R is the domain of f (i.e., projection of S onto xy -plane).

Example

Let S be the surface given by $z = \sqrt{x^2 + y^2}$, $z \leq 2$.

(a) Find the tangent plane to S at $(\sqrt{2}/2, \sqrt{2}/2, 1)$.

(b) Find the surface area of S .

Ans:

(a) $-\frac{\sqrt{2}}{2}x - \frac{\sqrt{2}}{2}y + z = 0$

(b) $4\sqrt{2}\pi$.